

Hongjie Xia

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EDUCATION

Boston University Ph.D. in Biology-Neurobiology, 2026

Northwestern University M.S. in Neurobiology, 2019

Boston University B.A. in Biology- Cell Biology, Molecular Biology & Genetics, 2018

RESEARCH EXPERIENCE

PhD Researcher – Systems Neuroscience | Advisor: Ben Scott

Boston University | Sep 2020 – Present

- Implemented and validated CAV/AAV-based chemogenetic targeting strategies in rat brain for cell-type-specific manipulation of noradrenergic neurons *in vivo*.
- Combined head-mounted two-photon calcium imaging with behavior to measure neural population dynamics during adaptive decision-making.
- Built reproducible Python workflows to preprocess and align neural and behavioral time-series data (1 kHz event streams) and analyze large-scale datasets (>500k trials).
- Developed computational models (drift diffusion models, GLM-HMM) to generate experimentally testable hypotheses linking circuit perturbation to behavioral policy changes.
- Performed stereotaxic viral delivery, perfusion, followed by immunohistochemical validation and quantitative fluorescence imaging to assess anatomical targeting specificity and labeling density using ImageJ and Python-based analysis pipelines.

Research Technician – Sensory Neuroscience | Advisor: Hojoon Lee

Northwestern University | Feb 2020 – Aug 2020

- Conducted conditioned taste aversion assays and performed perfusion, vibratome sectioning, immunohistochemistry, and confocal fluorescence imaging to quantify c-Fos-labeled neurons for circuit-level activity mapping.

Master's Researcher -Alzheimer's Disease | Advisor: William Klein

Northwestern University | Sep 2018 – Dec 2019

- Performed immunohistochemistry and quantitative fluorescence imaging to characterize A β and phosphorylated tau pathology in retinal degeneration models. Isolated and cultured retinal ganglion neurons to examine developmental and disease-associated phenotypes.

Undergraduate Researcher – Developmental Neurobiology | Advisor: Hengye Man

Boston University | Sep 2016 – Aug 2018

- Designed siRNA constructs and generated lentiviral vectors to knock down target proteins in primary neuronal cultures, followed by immunofluorescence imaging to assess protein localization and effects on neuronal differentiation.

PUBLICATIONS

- **Xia, H.**, Maheu, M., Kane, G., Scott, B.B. (2026). Regulation of the decision threshold by the locus coeruleus. *Neuropsychopharmacology*. <https://doi.org/10.1038/s41386-026-02399-x>
- Senne, R.A.*, **Xia, H.***, ...Scott, B.B.**, DePasquale, B.D.**. Diurnal rhythms of choice: a novel state-dependent drift diffusion model uncovers time-dependent changes in rat decision making. Under review.
- Frostig, H., Monasterio, A., **Xia, H.**, ... Scott, B.B. (2025). Three-photon population imaging of subcortical brain regions. *Nature Communications*. In press.
- Charkravarty, S., Delgado-Sallent, C., Kane, G.A., **Xia, H.**, ...Scott., B.B. (2024). A cross-species framework for investigating perceptual evidence. *eLife*.
- Bartley, S. C., Proctor, M. T., **Xia, H.**, ... Klein, W.L. (2022). An essential role for Alzheimer's-linked amyloid beta oligomers in neurodevelopment: transient expression of multiple proteoforms during retina histogenesis. *International journal of molecular sciences*.

PRESENTATIONS

COSYNE 2026 | Mar 2026

Xia, H., Maheu, M., Kane, G., Scott, B.B. Regulation of Decision Threshold by LC-NE. Selected poster presentation.

GRC Catecholamines | Aug 2025

Xia, H., Maheu, M., Kane, G., Scott, B.B. Regulation of Decision Threshold by LC-NE. Selected oral presentation.

SfN 2023 | Nov 2023

Xia, H., Rose, C., Kane, G., Li, S., Scott, B.B. LC-NE system improves performance in a visual evidence accumulation task. Program No. 168.28/OO14. Presented poster.

SfN 2022 | Nov 2022

Xia, H., Kane, G., Li, S., Scott, B.B. Low dose of clonidine improves task performance in a visual evidence accumulation task. Poster No.489.27. Presented poster.

TEACHING EXPERIENCE

Teaching Fellow | Boston University | Sep 2021 – Present

Courses: Principle of Neuroscience, Systems Physiology, Introductory Biology

- Led weekly laboratory or discussion sections, providing hands-on instruction in experimental design, physiological measurement, statistical analysis, and scientific writing. Developed problem-based learning materials to reinforce core concepts in cellular and systems neuroscience.

MENTORSHIP

Research Mentor | Boston University | 2023–Present

- Mentored multiple undergraduates and master's students in rodent handling/behaviors, histology, and data analysis.
- Supervised independent research projects for high school student (GROW outreach program).